

Ricardo Andrés Calvo Méndez

Ph.D student
Elmore Family School of Electrical and Computer Engineering
Purdue University
West Lafayette, IN 47907
Citizenship: Colombian (F-1 Visa)

☎ +1 (765) 595-1110
✉ rcalvome@purdue.edu
🌐 rcalvom.github.io
📄 ricardo-calvo

RESEARCH INTERESTS

My research focuses on the design and analysis of methods and tools that provide rigorous guarantees of software correctness. I am particularly interested in formal verification techniques, theoretical and practical foundations for reasoning about program behavior, and advanced approaches to code analysis.

My research interests include:

- **Formal verification:** Foundations and tools for establishing provably correct behavior in complex software systems.
- **Program analysis and testing:** Static, dynamic, and hybrid methods for understanding, validating, and rigorously evaluating code.
- **Optimization and efficiency:** Techniques for improving performance, reducing resource consumption, and identifying near-optimal implementations.

EDUCATION

Ph.D. in Electrical and Computer Engineering Fall 2025 – Fall 2030
Purdue University, West Lafayette, IN

B.Sc. in Systems and Computer Engineering Spring 2018 – Fall 2023
Universidad Nacional de Colombia (ranked #1 in Colombia), Bogotá D.C, Colombia

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2025 – Present
Purdue University — Advised by James Davis

- Serving as a researcher on *AutoUP*, an automatic unit-proof generation system for formal verification, contributing to its design and development.

Undergraduate Visiting Scholar (UREP-C) Feb 2023 – Jul 2023
Purdue University — Advised by James Davis

- Contributed to the design and implementation of *EmNetTest*, a framework for testing embedded network stacks.

PROFESSIONAL EXPERIENCE

Test Engineer & QA Analyst Sep 2024 – Jul 2025
Kravata SAS, Bogotá D.C, Colombia

- Led the testing area, advancing automation across end-to-end, integration, and unit testing while implementing CI/CD pipelines within a DevOps environment.

Fullstack Software Developer Aug 2023 – Aug 2024
Fit Ideas SAS, Bogotá D.C, Colombia

- Led the development team in the strategic planning, management, and deployment of mobile and web applications.

- Supported the implementation of custom functionalities as a developer, and contributed to the design and management of relational database systems.

REFEREED CONFERENCE PUBLICATIONS

- [C-1] P. Amusuo, D. Liu, R.A.C. Méndez, J. Metzman, O. Chang, and J.C. Davis. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE) (FSE-Industry'26). 39% acceptance rate (66/171). 5 pages. *Presenting author*.
- [C-2] H. Peng, A. Zhong, R.A.C. Méndez, K.G. Kalu, and J.C. Davis. *How Do Agents Perform Code Optimization? An Empirical Study*. Proceedings of the 23rd International Mining Software Repositories Conference — Mining Challenge track (MSR-Mining'26). 53% acceptance rate (62/116). 5 pages. *Presenting author*.
- [C-3] P.C. Amusuo, R.A.C. Méndez, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23). 21% acceptance rate (134/629). 13 pages.

ACADEMIC COMMUNITY SERVICE

Volunteer Service

Student volunteer, ICSE 2026 <i>Rio de Janeiro, Brazil</i>	2026
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Reviewing Service

Reviewer, MSR 2026 — Mining Challenge Track	2026
Sub-reviewer, USENIX 2026 — Cycle 2	2026
Sub-reviewer, ICSE'26	2026

CONFERENCE ATTENDANCE

39th IEEE International Conference on Software Maintenance and Evolution (ICSME 2023) <i>Bogotá, Colombia</i>	October 2023
ACM/IEEE 48th International Conference on Software Engineering (ICSE 2026) <i>Rio de Janeiro, Brazil</i>	April 2026

EXTERNAL GRANTS

- [G-1] **Travel Support for the ICSE 2026 Student Mentoring Workshop (SMeW)**
National Science Foundation (NSF) and ACM SIGSOFT
Rio de Janeiro, Brazil. April 2026.
- [G-2] **“Ser Pilo Paga” Scholarship**
Ministry of Education of Colombia
National merit-based scholarship awarded for top performance on a national standardized high-school examination, with 100% funding for undergraduate tuition and academic expenses.